

Science Applications

Achieving breakthrough science requires brilliant people, dedication, and the proper laboratories and tools. The computer has become an essential, sophisticated laboratory tool that provides capabilities that were not possible before. One thing the computer does is enable the scientist to manage huge volumes of data. Scientists acquire data through observations, often using instruments such as radio telescopes, electron microscopes, etc., as well as by synthesizing data by computational modeling of natural phenomena that result in highly refined simulations.

A major problem that confronts the scientist is how to make sense of the billions of numbers provided by these data sources. Hence, the field of scientific visualization was born in which scientific data is represented as visual images for the purpose of data analysis and presentation. Virtual reality is also a powerful tool in enabling scientists to explore the world via related techniques such as telepresence. Telepresence allows a scientist to be virtually present in places that are not normally possible because they are too dangerous, too far away, too small, too large, or otherwise not conducive to human visitation.

Many of the phenomena that scientists study are three-dimensional (or more) in nature. Thus, it is a natural outgrowth of the field of scientific visualization to take advantage of the three-dimensional nature of virtual reality display and interaction techniques.

4.1 AREAS OF APPLICATION

Scientists have applied virtual reality to scientific visualization in a number of fields. Computational fluid dynamics allows the scientist to simulate flow under various conditions. The ability to visualize flow allows the scientist to better understand the fluid's interaction with objects in the environment

(such as the flow of air over a wing, or blood through an artery). Because most flow is three dimensional, the three-dimensional virtual interface is often advantageous.

The buildings we live in are three dimensional in nature. Understanding their structural properties allows engineers to design and build safer buildings that are able to stand under heavy wind conditions, earthquakes, etc. Finite element models of structures, beams, automobile fenders, etc., can be visualized in virtual reality and allow the scientists to understand their behavior under varying conditions. Chemists, biochemists, and others are interested in how atoms and molecules interact. The ability to visualize those interactions is a powerful tool in understanding basic reactions, drug design, properties of materials, and much more.

The ideas discussed above are but the tip of the iceberg in current and future uses for scientific visualization in virtual reality environments.

4.1.1 Exploration

Many areas of science require the scientist to leave their laboratory and go into the field for observation, data collection, experimentation, and other duties. To do so sometimes involves considerable expense, danger, time, and inconvenience. Consider space exploration. NASA has spent billions of dollars, as well as the cost of lost lives, to allow scientists to gather data from space. However, the benefits from such exploration have been immense. Virtual reality and related technologies such as telepresence are allowing scientists to explore remote areas from the safety and comfort of their own laboratories. These technologies are also allowing scientists to explore areas that have never been explored before, such as the bottom of the ocean, within nuclear waste dumps, and other areas that are hostile to human life.

This chapter illustrates the use of telepresence as an aid in exploring under the ocean, as well as the surface of Mars.

4.1.2 Physical system simulation and interaction

In many cases it is advantageous to be able to visualize and manipulate objects in a simulated or real system. For example, in the *GRIP* application described later in this chapter scientists are able to reach out and physically manipulate representations of molecules, and observe visually their interactions. In addition, the scientist is able to feel the forces between molecules via a haptic interface. Another example of physical simulation and interaction is the Air Force solar system modeler, which allows scientists and engineers to observe locations of planets and other celestial objects, as well as satellites and other man-made objects from various perspectives, and at various times in the past and in the future.

There are many overlaps between applications in this chapter and applications discussed in other chapters of this book. For example, the scientific visualization application *Crumbs* has found many applications in the medical field due to the reliance of physicians on image data from three-dimensional objects. In the field of virtual prototyping, additional

information can be overlaid on the models to visualize concerns such as stress, strain, temperature hot spots, etc. In much the same way that scientific visualization tools educate the scientist in matters of what data indicates, these same tools can be used for educating students about scientific phenomena. Finally, these same tools may find use in the fields of anthropology, archeology, and any study of natural phenomena, such as Sandia's RAPTOR study, detailed in Chapter 7.

Another path for visualization in virtual reality is to enable current visualization tools to exploit the new capabilities of virtual reality. This allows scientists who are already familiar with visualizing their data using common software packages to take advantage of the immersive VR environment. This process includes packages such as *AVS*, *COVISE*, and *SciRun* adapted to *CAVE* environments.

Finally, as computational power increases, existing bottlenecks in simulation and visualization will be relieved. For example, Professor Eric Loth extended the notion of computational fluid dynamics visualization in *CAVE* environments to also support multiphasic conditions. Virtual reality has already demonstrated its utility in the scientific process, and new developments will allow scientists to use VR to aid them in understanding and explaining more areas of difficult scientific problems.

Scientists are often among the early adopters of applying new technology in their work. Virtual reality is no exception. Many early examples of applied VR come from the labs of NASA, where researchers began exploring it as a means for space exploration and visualization in the 1980s. Whereas NASA had to develop the technology as well as the applications, more recently many more scientists have been able to experiment with VR by using established VR hardware and software.

The primary goal of science is to explore natural phenomena around us, and derive explanations of what is found. We often refer to the originations of new explanations as *insight*. The use of various media (sketches, written word, x-ray photography, and now virtual reality) has played an important role in the process of gaining insight.

4.1.3 Areas of application in science

Scientists have used virtual reality to interface with their work in three primary ways: the visualization of observed and simulated data, exploration of places that are difficult to travel to, and direct interaction with simulations of physical systems.

- Scientific visualization
- Scientific exploration
- Physical system simulation and interaction

The applications presented here cover these application areas in whole or in part, and give insight into the possibility of virtual reality for scientific applications.

CASE STUDY 4.1: VIRTUAL WINDTUNNEL

Executive Summary

One of the earliest scientific uses of virtual reality was the application from NASA Ames known as the *Virtual Windtunnel*. The *Virtual Windtunnel* portrays many interesting ideas that persist to this day, which is remarkable since the application designers had virtually no predecessors to emulate. The *Virtual Windtunnel* is used in a head-mounted display with a data glove, a *BOOM* display, and in projection-based displays.

The *Virtual Windtunnel* allows scientists and others to experiment with computed flow fields in a virtual environment that allows an actual wind tunnel to be emulated, as well as allowing additional capabilities not found in typical wind tunnels. The *Virtual Windtunnel* emulates many common tools used in an actual wind tunnel, such as allowing the scientist to release “smoke” in the flow field. In an actual wind tunnel the act of introducing smoke affects the flow field. In the simulated environment, however, the scientist can choose to have their tools affect the flow field or not. The *Virtual Windtunnel* application provided an example to others of the capabilities of virtual reality for scientific discovery and explanation. Although the application was built in the laboratory, many versions of the application have been demonstrated in numerous venues allowing others, including the general public, to learn more about computational fluid dynamics, fluid flows, and virtual reality.

Introduction

A critical problem for scientists and engineers is to understand the unsteady flow of fluids. In particular, it is of great importance to analyze the effects generated as a solid object passes through some fluid medium, or vice versa. One means of understanding these flows is through computational modeling. However, computational models of complex systems generate too much data to be analyzed as individual numbers. Instead, scientific visualization is often used to aid in the analysis process.

The task of visualizing the output of computational simulations in three dimensions is itself a nontrivial problem. The *Virtual Windtunnel* from the NASA Ames Research Center is a system that addresses the problem of visualizing the

results of such complex three dimensional simulations using virtual reality. The *Virtual Windtunnel* allows a researcher to interactively explore flow fields in a three-dimensional virtual world using common visualization techniques.

Steve Bryson and Creon Levit began work on The *Virtual Windtunnel* application in 1989. The concept of the visualization tool came out of the NASA Ames Virtual Interface Environment Workstation (View) Lab (Michael McGreevey and Scott Fisher) working with physicist Creon Levit as a project to establish the potential benefits of the medium of virtual reality. The project has continued to evolve, and in 1998 was released on the Internet for others to use.

Application Description

The *Virtual Windtunnel* application was a pioneering virtual reality application in the field of fluid dynamics. Prior to the existence of the *Virtual Windtunnel*, experiments such as studying the airflow over a wing were carried out in actual wind tunnels. In an actual wind tunnel, techniques such as releasing smoke into the wind tunnel allowed scientists to see the flow of air over the wing. The *Virtual Windtunnel* allowed scientists to see flows over a virtual wing using scientific visualization techniques such as particles and streamlines. The *Virtual Windtunnel* was primarily experienced using a *BOOM* device.

The *Virtual Windtunnel* was also made into a desktop application for those scenarios that did not require VR hardware, and for times when VR equipment was not available.

Representation of the Virtual World

In the immersive display systems the virtual world consists of any objects involved in the computational experiment (such as an airplane wing, or the body of the space shuttle) as well as the flow field and its representations. There are a variety of tools that allow the user to visualize the flow with some graphical metaphor to a real-world equivalent representation. For example, as in the real world, “smoke” can be released into the virtual world to view the flow from an arbitrary location. Unlike the real world, the presence of measuring tools does not affect the flow. The ability to measure without interference is very useful in exploring sensitive areas such as boundary layers and chaotic regions.

The visualization tools are derived from common idioms used in many computational fluid dynamics (CFD) visualization applications. Idioms include vector data fields represented by streaklines, particle paths, streamlines, and material tracing. Scalar data can also be visualized with techniques such as global or local isosurfacing, pseudo-color mapping, and probing the data to view the underlying numeric values [Sherman et al. 1997].

The location of the gloved hand or handheld pointer is represented in the scene as a 3D crosshair marker. This avatar allows the user to get a better sense of the position of their hand relative to the other objects in the world. Only visual information is displayed to the participant. Users of the tool have not expressed a significant interest in other sensory modalities such as sound or haptic feedback. Therefore, no effort has been made to integrate other sensory rendering/display into the package.

Interaction with the Virtual World

Each tool has a set of “grab points” that allow the user to do different things. Depending on which “point” is grabbed, the user can “grab-and-move” in order to move the source of the visualization to a new location, or resize or otherwise adjust the tool. The display is continually updated to reflect the flow in the new region. There are a variety of visualization techniques, and display options that the user can control.

Travel is performed by a method common to all the interface devices that this application has been used in the “world-in-hand” paradigm. In both the *BOOM* and the *Responsive WorkBench (RWB)*, the user can grab the world to move and orient it in any direction, allowing them to view the world as they desire. Both display types also allow for some form of physical locomotion to adjust the user’s perspective on the world, but with some limitations imposed by the nature of the displays.

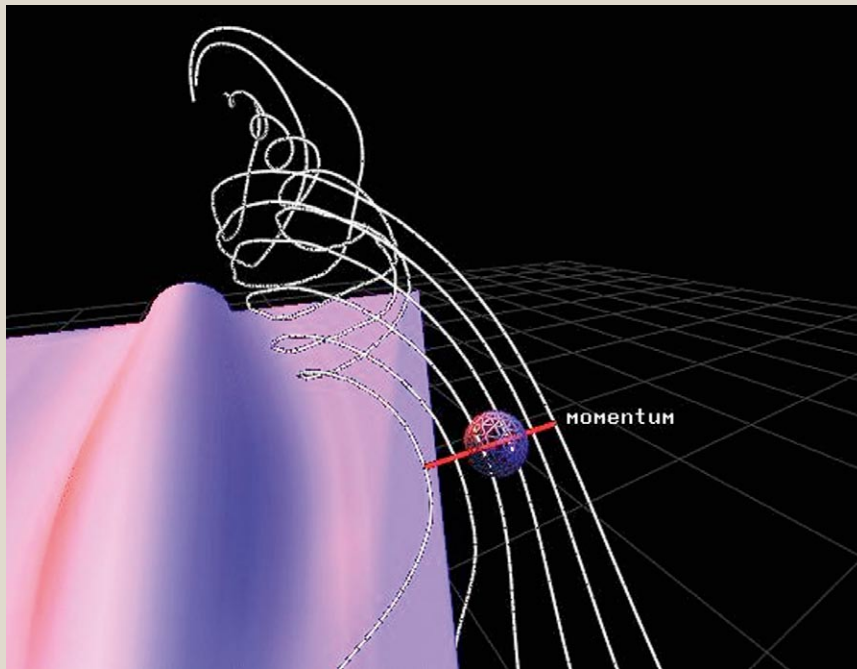
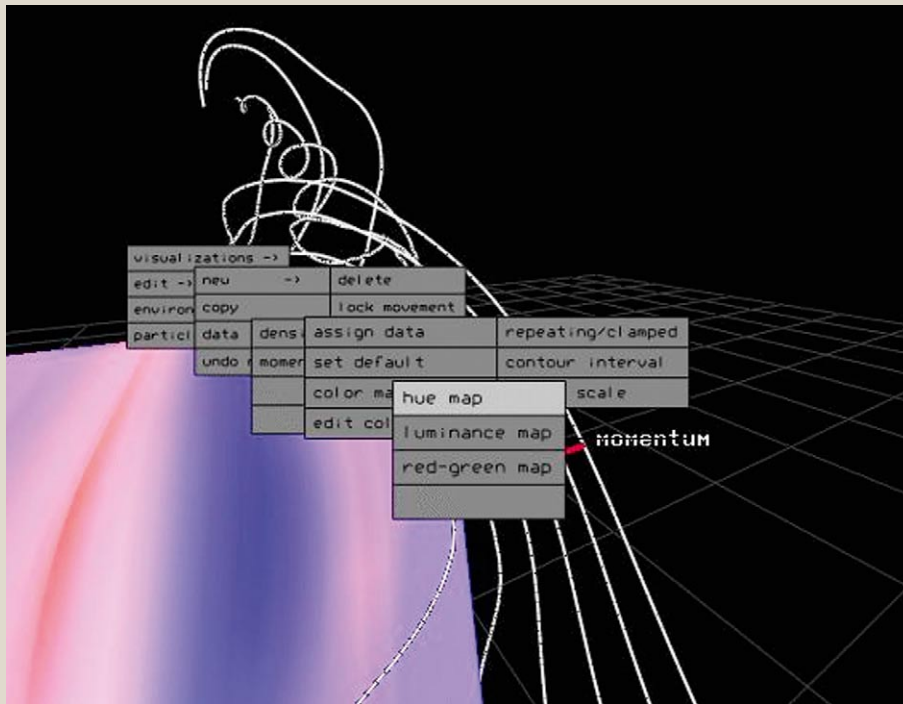


FIGURE 4-1

This example visualization shows streaklines over an airfoil. The control bar allows the user to change the location by grabbing the ball, or to affect the number of streaks by stretching or contracting the bar.

**FIGURE 4-2**

The Virtual Windtunnel provides hierarchical, pop-up menus to allow users to select from various options.

The *BOOM* display allows users to physically walk through the data, and turn their head in any direction, but has limited reach. The mechanical linkages of the *BOOM* also limit the user from walking to certain locations. When using the *RWB* display the user cannot walk around to all sides of the table or through the table, and consequently the virtual world. Thus, physical locomotion must be supplemented by other travel techniques. The “world-in-hand” interface paradigm accommodates both types of displays and allows the user to view the world from any relative position. To activate world-in-hand travel in the *RWB*, the user points to open space, presses the pointer button to “grab” the world, and moves the pointer to cause the same relative movement of the virtual world. Both the *BOOM* and *RWB* version provide controls for the users to change the relative scale between themselves and the world, as well as move and manipulate the visualization tools.

In addition to physical locomotion and world-in-hand travel, some demonstration versions of the *Virtual Windtunnel* also

provided the point-to-fly travel paradigm, which may be more familiar to participants at the particular demonstration venue. In particular, the *BOOM/glove* system allowed the user to travel by putting their gloved hand into the pointing posture and guiding the direction of travel by pointing to their desired destination.

A user can also control the visualization parameters. This interaction is done from within the virtual world via hierarchical pop-up menus. Menus are invoked by making a “pointing” posture at empty space with the gloved hand. They then select an item by pointing their index finger to their choice and releasing the hand-posture. The menus are hierarchical to better organize the set of choices.

Venue

Although the *Virtual Windtunnel* was originally designed in a laboratory setting, the application has been demonstrated in many venues using a variety of technologies. Many people

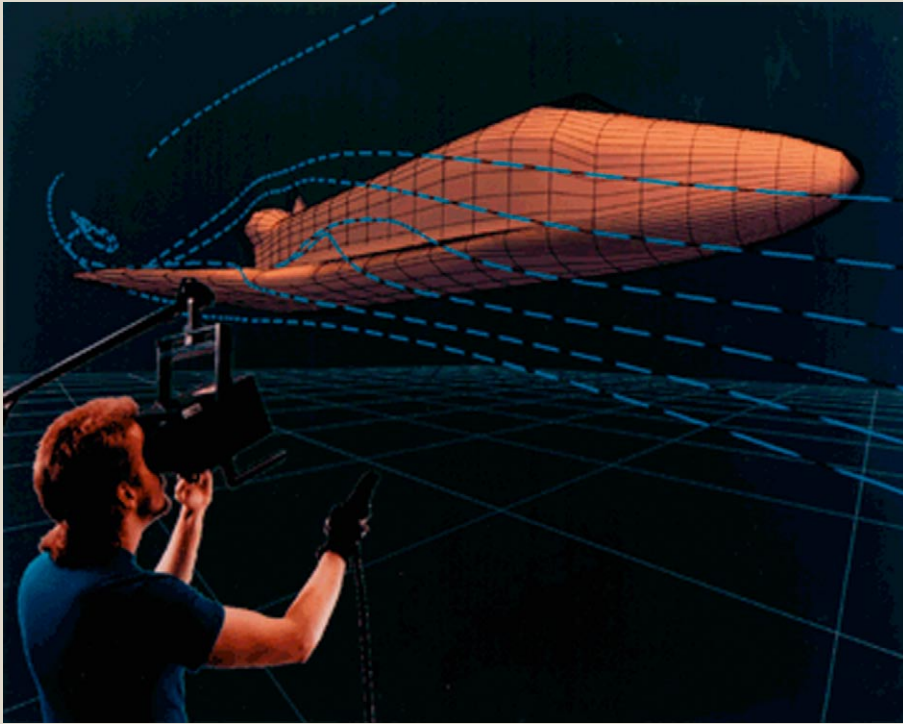


FIGURE 4-3

A view of Virtual Windtunnel creator Steve Bryson using a BOOM to view streamlines over the wing of a space shuttle. The upper portion of the photo shows the image Bryson is currently viewing. Bryson uses a DataGlove to enter commands via hand gestures.

gained their first exposure to virtual reality by witnessing the *Virtual Windtunnel* at conferences and other public settings. The application has been experienced by numerous people, from rocket scientists to first-graders. In addition to its originating lab at NASA-Ames, *Virtual Windtunnel* has been ported to different display technologies in other laboratories such as the *Responsive WorkBench*. Additionally, researchers at Brown University have created a separate user interface.

VR System

The initial version of the *Virtual Windtunnel* used a BOOM display device, a stereoscopic, head-tracked viewing device. A glove was worn to allow the user to interact with the system by issuing commands via hand postures. Parameters of the visualization tools were entered via a traditional keyboard and mouse. Use of the BOOM-style VR display was

convenient because it still allowed the user access to the computer keyboard.

With little time involved in “suiting up,” transitioning between the VR display and the keyboard is quick and easy. The user can remove their head from the BOOM, use the traditional workstation for manipulating selected parameters, and then bring the BOOM back to their face.

Later, the *Virtual Windtunnel* was equipped to handle a variety of input and output devices such as the RWB tabletop projection system.¹

¹or the *Immersive WorkBench*, as the version marketed by Fakespace Inc. is called.

In the *WorkBench* version, a handheld pointer device with a single button replaces the glove input. This changes the interface somewhat in that there are less direct control inputs available to the user because they can only push a single button. With the *BOOM*/glove system the user can generate multiple hand postures and still use the buttons mounted on the handle of the *BOOM* display, allowing the operator more flexibility.

The *Virtual Windtunnel* also gives the option of using a desktop monitor and mouse interface for situations when it is impractical, or unnecessary, to utilize the VR displays. Because the application was designed to be interactive, many people have found benefits in using the *Virtual Windtunnel* even as a desktop application.

Application Implementation

Though commonly used to examine precomputed datasets stored locally on the graphics workstation, the application developers extended the capabilities of the application by providing an interface that enables the simulation computations to be calculated on a remote supercomputer, and delivered to The *Virtual Windtunnel* via a high-speed network. The first step was to introduce an asynchronous process to produce the visualization representations, apart from the rendering of the visualization. This separation allowed the user to continue interacting with the virtual world in a smooth, continuous manner.

Access to greater computational resources enabled the exploration of time-varying simulations. However, expanding from the visualization of static steady-state flow simulations to time-varying simulations led to a significantly more complicated data management issue. There are a variety of time-management issues that arise in applications that allow the user to control simulation time. Recent implementations of the *Virtual Windtunnel* address many of these issues in interesting ways [Bryson and Johan].

A particularly interesting feature of The *Virtual Windtunnel* is the use of a “computational budget” to maintain high frame rates of both the graphical rendering and the computation of subsequent time steps. Bryson and Johan assert: “Many researchers have indicated that when interactively exploring a data set they would rather have a fast, less accurate answer than a slow accurate answer.”

Thus, for example, if a large number of particles are used to show the flow in some region, the computation of the particle locations cannot be done quickly enough with a more complex algorithm (e.g., fourth-order Runge-Kutta). In these cases, computation switches to a less accurate but faster algorithm (e.g., second-order Runge-Kutta) until there are enough resources to reinstate the more accurate algorithm.

Related Applications and Follow-on Work

We use the term “Virtual Windtunnel” in this chapter to refer to the specific virtual reality application. However, many programs and systems bear the same moniker. One system by Denford Ltd of West Yorkshire England provides CFD simulation and visualization for vehicles and is aimed at a wide audience of general users. Though the creators advertise the product with the term “virtual reality,” the application does not involve any tracking, stereo, or kinesthetic input and therefore does not fit the genre of virtual reality as we define it in this work. Other “virtual windtunnels” include freely available simple Java programs, interfaces through Web pages, and specific applications of Flovent software (made by The Mechanical Analysis Division of Mentor Graphics Corporation, formerly Flomerics).

The obvious application that (Bryson and Johan’s) *Virtual Windtunnel* addresses is the need to visualize computational fluid dynamics data, and the adoption of the “Virtual Windtunnel” name by several related applications testifies to this fact. Perhaps a most illustrative example of a fully VR implementation to merge the CFD and visualization worlds is *VE-Suite*. Originally developed at Iowa State University by the Virtual Engineering Research Group under the direction of Professor Mark Bryden., *VE-Suite* is now available in open-source format under the GNU Lesser General Public License.

VE-Suite focuses on allowing the user to simultaneously generate engineering simulations and visualize the resulting data. The integration of these two often-disconnected actions allows fast design iterations and real-time decision making. Bryden states,

“VE-Suite is a tool to develop complete virtual environments that include graphics, physics, and

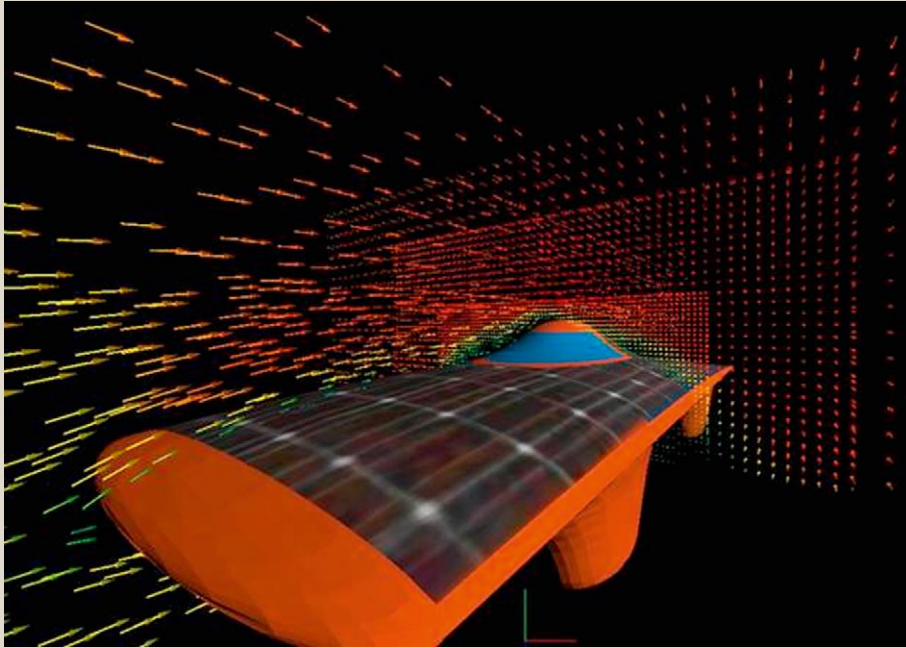


FIGURE 4-4

VE-Suite provides a multifaceted interface for visualization of computational fluid dynamics data as well as several other engineering applications.

easily understood user interactions. It can be used across many display hardware configurations, from desktop to the largest multiwall stereo displays. VE-Suite is currently used in a number of companies, national labs, and universities. VE-Suite has been used on applications ranging from finance, engineering, biology, processing plants, and materials science. In 2006 VE-Suite won a R&D 100 award for being one of the top 100 R&D developments of the year."

Conclusion

The *Virtual Windtunnel* is an example of a virtual reality application that is based on a complex simulation of natural phenomena. Many of the "virtual" techniques implemented in the application have their origins in real-world measurement techniques. In the virtual world, however,

there are fewer constraints on the number and location of tools since the "virtual" tools don't interfere with the experiment.

Most of the equipment used during the project was also shared with other efforts, especially the large Silicon Graphics workstation computers. The budget for this project varied, but was approximately one million dollars spent over the course of the first 7 years.

The *Virtual Windtunnel* was a successful VR application on many levels. First, it was an early demonstration of a scholarly use of the medium of virtual reality. Second, it provided some users within NASA a tool that helped them gain new insights about their data. Finally, it led to interaction techniques that were found to be useful even on a nonimmersive desktop system.

**FIGURE 4-5**

Mark Bryden and Steve Gent use VE-Suite to analyze the airflow through the fan blade of an engine.

Prior to release to the general public, there were few users at NASA and elsewhere that worked with various versions of the application. Feedback from these users was generally positive. Many users reported that they like the ability to directly manipulate the visualization tools to view their data, and some reported that they found new features in datasets with which they already had considerable experience viewing via other technologies. The release of the code to the public was done in order to gain a wider audience of application.

In addition to cofounding programmers Bryson and Levit, other significant contributions have been made by Sandy Johan, Brian Green, David Whitney, Leslie Schlecht, David Kenwright, David Kao, and Tom Meyer.

Steve Bryson summarizes the benefits he sees in virtual reality as demonstrated by the *Virtual Windtunnel* project:

The virtual windtunnel presents abstract phenomena with a strong 3D presence and allows you to explore that phenomena in a completely intuitive manner. We've found that from first grade students to seasoned CFD researchers, people reach a quicker understanding and have a stronger memory of the abstract phenomena through intuitive exploration combined with a strong 3D presentation. I think this is the power of virtual reality: by hiding the computer interface, total intuitive and natural focus on the virtual world provides a quicker and more compelling grasp of what's happening in that world.

CASE STUDY 4.2: ARMY CORPS OF ENGINEERS

BayWalk

The Army Corps of Engineers Waterways Experimentation Station (CEWES), in Vicksburg, Mississippi, completed an extensive study using VR to visualize the Chesapeake Bay. CEWES collaborated with the National Center for Supercomputing Applications (NCSA) to produce another VR application for visualizing the bay. This project served several purposes. One of the primary purposes was as a technology transfer project. CEWES had acquired an *ImmersaDesk* and was interested in doing a collaborative project with NCSA to gain experience in applying it to scientific applications. Of several possible projects, NCSA was especially attracted to the Chesapeake Bay project due to the importance of the bay, as well as the fact that the simulation of the bay was computationally complex and highly multivariate.

Application Description

The *BayWalk* application was created primarily as a scientific visualization and collaboration tool for exploring the results of a complex computational science simulation. In the computational model, the three-dimensional bay was divided into 4000 volumetric cells or “bricks.” The simulation calculates flow information and the presence of certain substances throughout the bay over time. There are many variables at each cell in the simulation. However, for the initial study only three were visualized—*chlorophyll*, *salinity*, and *dissolved oxygen*. More recently, two more variables were added—*nitrogen* and *phosphorus*. (The latter two are of particular interest because they are contained in fertilizer, one of the sources of ecological concern.) The computational model has been calibrated to match observed data by modifying coefficients and other computational parameters so that the simulation results based on known initial conditions match the actual results observed in nature.

Representation in BayWalk

As with most virtual reality experiences, *BayWalk* is represented primarily visually. To begin with, the participant sees a planar representation of the bay with text markings and symbols to give landmark locations, including city and river names. On command from menus controlled by the wand, the participant can turn on and off a variety of visual representations, including the computational grid, glyphs,

cutting planes, and volumetric idioms. These representations are very useful for the scientist, and given some explanation, can also aid the nonexpert viewer in understanding how the computational model works.

One of the visualization tools available is a display consisting of “glyphs” that indicate the scalar values of some subset of the data within each cell. In the original version, the three variables were mapped to a three-vector axes where the length of each axis indicated the amount of that chemical. Chlorophyll was mapped to the green vector, salinity to magenta, and dissolved oxygen to white. These glyphs enable the researcher to see relationships between the variables as the simulation proceeds over time.

One modification was to add two new variables (amount of nitrogen and phosphorus-limiting potential) to the glyph display. The glyph was modified to show all five variables on a five-pronged display. The nitrogen-limiting potential is represented as blue and phosphorus red.

Over the change of seasons one can observe that in the spring runoff, high levels of nutrients are carried into the bay generating an algae bloom (indicated by an increase in chlorophyll). Later in the season the algae dies and falls to the bottom of the bay. It decomposes there, consuming oxygen in the process. This is evident from the shortening white vectors in the depths of the bay. The researchers indicated that this tool has proven effective, and even non-scientists have been able to understand the dynamics of these environmental interactions. By including the phosphorus and nitrogen-limiting potentials, the viewer can see that over the course of a year there is a seasonal switch from phosphorus to nitrogen. During the spring algae bloom, the system is phosphorus-limited, whereas during the period of summer maximum productivity, the system is nitrogen-limited. Another visualization tool available in the system is an “orthogonal” slicing plane—orthogonal with respect to how the data are stored. The slicing plane is a two-dimensional surface that cuts through the data. It displays a color mapping based on the value of the chosen variable at each point the plane intersects. The data grid is irregularly shaped, but it is roughly aligned to and regularly spaced across the Cartesian axes. The plane can be

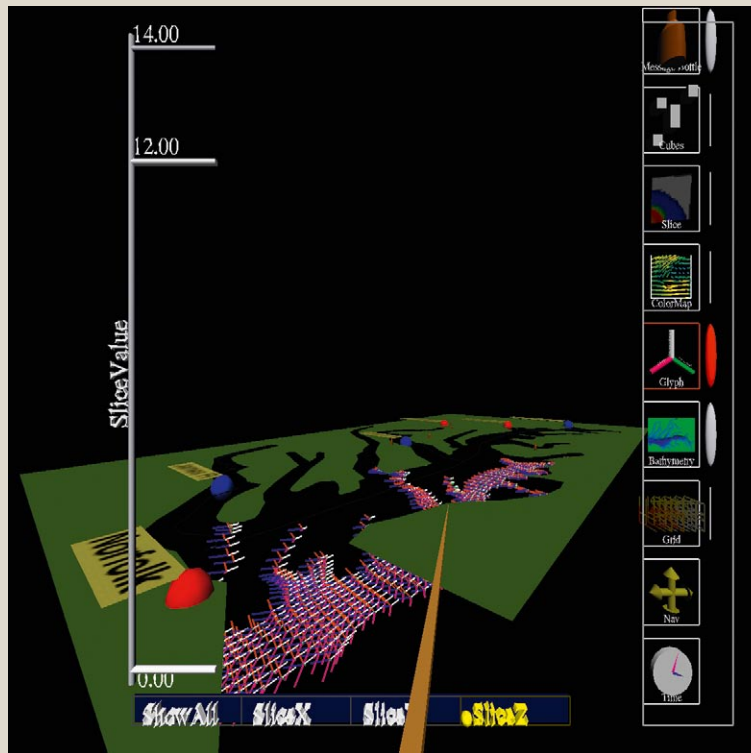


FIGURE 4-6

Abstract glyphs are used to show the concentrations of oxygen, chlorophyll, and salinity. The length of each axis of the glyph represents the amount of each substance. The color indicates the substance.

translated along any of the Cartesian axes, and a submenu provides a selection of data values that can be represented (Salinity, chlorophyll concentrations, etc.).

A simple crude isosurface can be generated by selecting a tool that colors all the bricks that fall within a user-selected range of data values.

Travel and other Interaction in BayWalk

The handheld wand is used by the participant to travel through the virtual space, travel through time, and select menu items. Travel is by pointer-directed fly-through, allowing the user to easily control movement throughout the entire data space. All other interactions are performed through a menu system.

The menu system is context sensitive and displays a list of choices based on the current interaction mode. The main menu (for toggling visual representations and setting the current mode) can be activated either by holding the wand above it (useful on the *ImmersaDesk* display), or when the wand is pointing at it (more useful for Wall or *CAVE*-projected displays). Once the main menu is activated, the pressure joystick on the wand is used to slide choices up and down the menu, and to stop on the desired selection. Certain selections may invoke other controls, such as submenus, or a slider.

The interface limits the user to control only one thing at a time. More importantly, this co-use of moving the wand and its joystick both for travel and for menu interactions

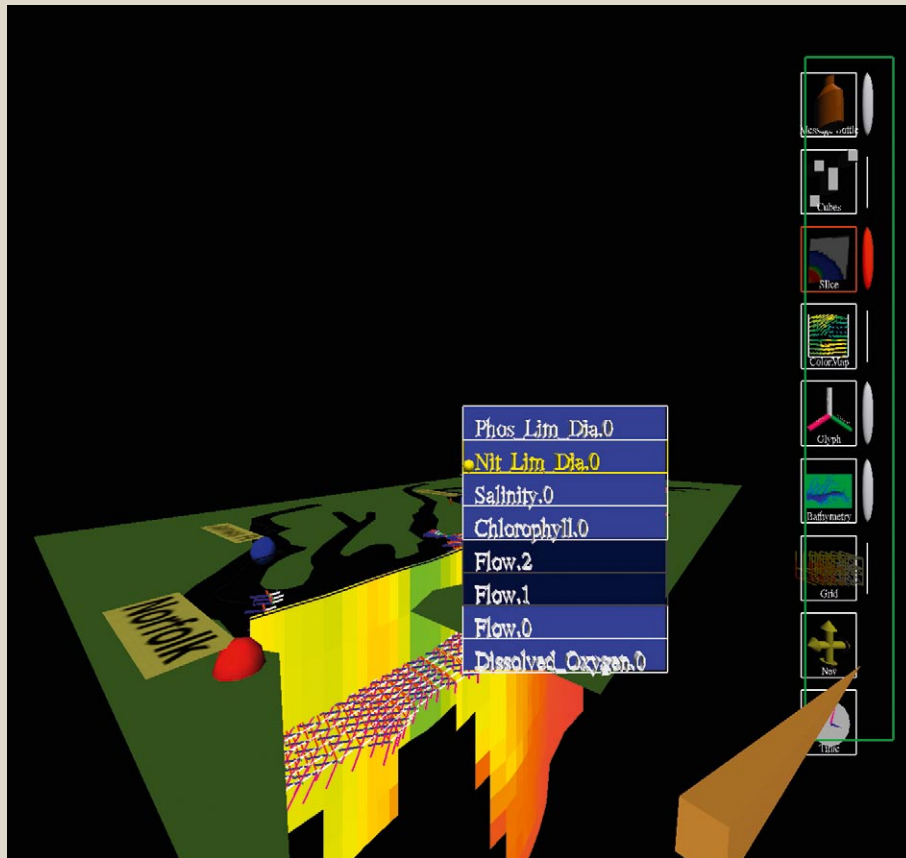


FIGURE 4-7

A slicing plane is texture mapped with colors that represent the chosen variable.

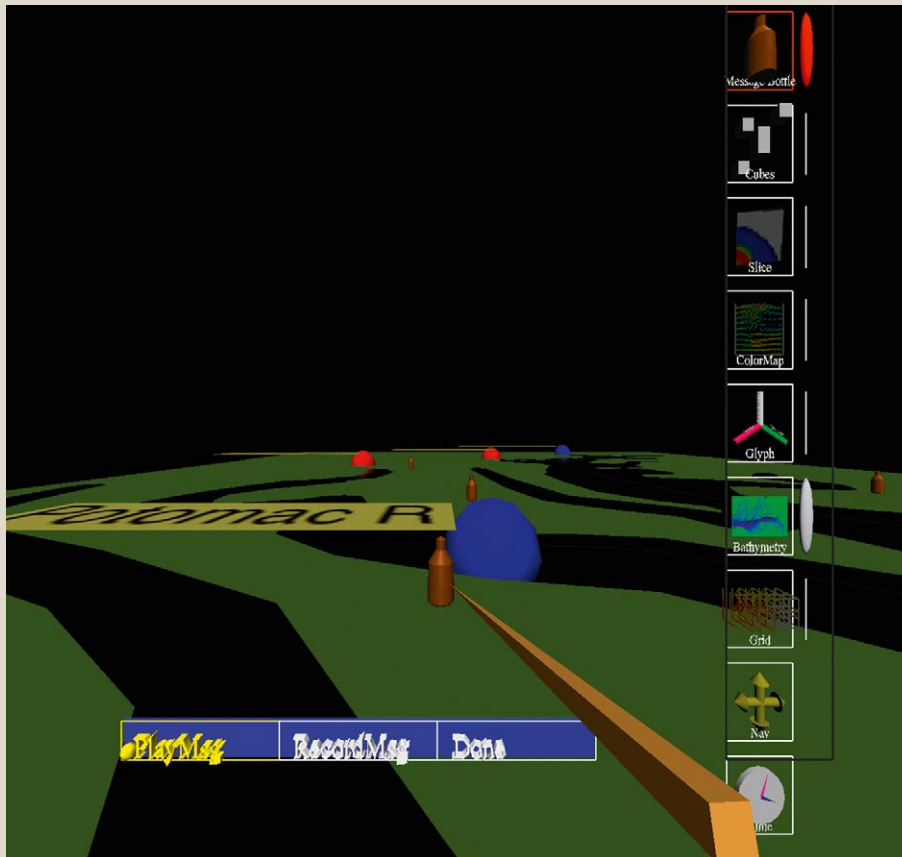
can cause interference and confusion. As the user travels through the space, they can very easily activate the menu system as they move the wand to specify direction. This accidental activation of the menu causes them not only to stop moving, but also to change the menu selection, since the pressure they are applying to the joystick now affects the menu.

A slider interface appears on the screen when specific menu selections are made. The slider is used for moving a cutting plane throughout the data, or to manipulate time. The dataset used in the original version covers a period of one year allowing the user to analyze the annual stages

of the ecosystem. The slider, located at the left side of the main viewing surface (e.g., front wall of the *CAVE*) is used to select the time value for which simulated data to use. Animation of the data over time can only be done by manually sliding the time control. Thus, there is no way to put time in motion, and simultaneously manipulate other features or navigate through the data as time advances.

Collaboration in BayWalk

Because of interest in the bay by geographically diverse researchers, this VR experience was created as a collaborative tool that can be used by multiple participants in

**FIGURE 4-8**

In the BayWalk application, the main menu is vertically oriented, whereas submenus related to the chosen menu appear horizontally.

separate geographic locations. It was designed to allow any number of participants, but was tested with only two.

Each user is represented in the environment by a fish-shaped avatar. The position and orientation of each user's wand is represented by a pointer emanating from the fish. Combined with an audio communication channel, collaborators can have discussions and observe where other participants are looking and pointing.

In a collaborative session, a multicast connection is usually established. The multicast session appears on an

ImmersaDesk and allows the participants to see and hear each other as well as the surrounding environment at the remote location. The multicast connection has proven especially valuable when demonstrating the system to observers who are unfamiliar with the system. Even if appearing in a separate window on the screen, seeing the remote collaborator and their lab makes it more obvious that collaboration is taking place, and that the fish is a representation of what the remote users are doing in the virtual space. The video feedback also aids the collaborators in understanding what is happening at the other end.



FIGURE 4-9

The passage of time in the BayWalk application is depicted by a vertical timeline. The timeline can also be used to set the view to a particular time.

For example, the local user can see when other people enter the collaborator's lab and call them away from the system. If the other participant takes off their glasses and lays them upside down on a table, without the video/audio connection, it would look as though that person's fish went "belly up" with no explanation.

In addition to collaborating in a live two-way communication session, some experiments have been done to allow researchers to collaborate by leaving annotations in the *BayWalk* world. Messages can be deposited in the virtual world in the form of voice annotations. The voice messages are left using the metaphor of a "message in a bottle," that is, wherever an annotation is left a bottle icon is placed in the world. By pointing at the bottle and pressing a button, other researchers can access the information. The same

researcher may even access the annotation in a future interactive session.

The message in the bottle paradigm enables a certain amount of asynchronous collaboration to take place. In the initial instantiation, the annotations do not persist across sessions. To reduce bandwidth requirements for collaborative data exploration, each site has a copy of the computational data on their local computer. Thus, the only data transmitted between sites is user positions with the positions of their wands, and if utilized, the multicast communication and voice messages.

VR System

This application was written using the *SGI Performer* (2.0) library and C++. The *CAVE* library handles the multiple-screen display, tracking, wand events, sharing user

information between sites, and other virtual reality aspects of the interaction. The initial version, demonstrated publicly at the Supercomputing '96 conference held in Pittsburgh, took about 6 months to complete.

Initially, it took approximately 8 hours on a Cray Y-MP supercomputer to run a simulation encompassing 1 year. This is not fast enough for the researchers who had a goal of doing a 50-year simulation. Thus, the VR application was run as a post process to the supercomputer simulation. The simulation results were computed and stored, and could then be explored using a variety of tools at whatever time scale the researchers desire. Later, the code was optimized to calculate a 20-year simulation in less than a real day, a 50-fold speedup. The researchers later moved the program to a Cray T3E. Running on 64 processors, this platform could simulate 20 years in less than a day on a 10,000-cell grid.

Originally designed for the *ImmersaDesk*, the application also runs in the *CAVE* environment. Because the application was specifically designed for the *ImmersaDesk*, which has an angled viewing surface, some user interface elements required minor adjustment to display correctly on the other devices. The *CAVE* library also supported display in head-mounted displays.

The application authors originally planned to enhance the ability to interactively probe a region of interest in the bay in high detail. This capability would more fully exploit the capabilities of a VR system by offering an “inside out” view of the data in addition to the “outside in” view that is how the current system is typically used. Without the ability to probe the dataset in detail at a given place, it is less relevant to know where your collaborator is positioned in the virtual environment. When allowed to probe into the data, it then becomes interesting to see what region the collaborator is exploring.

A possible enhancement to the synchronous collaboration interface that was considered was to duplicate control information between sites, so one user could manipulate how all sites see the visualization. Such a capability would allow a scientist who is trying to demonstrate a feature or an interesting aspect of the data to those at another site to show them directly by controlling their movement and

tools, rather than have to explain exactly how to set all the controls and requiring them to make those adjustments. Changing to a persistent storage of annotations between collaborative sessions would also enhance the usefulness as a collaborative tool.

Other than the multicast session and the voice messages, there is no audio used in this application. There are no immediate plans for using audio to provide sonification, but some feedback for menu interaction, etc., is desirable. Voice input is also a desired feature. With voice input, much of the problems caused by interference of the menu with the travel interface can be reduced.

Related Applications and Follow-on Work

One of the first applications of virtual reality to the visualization of oceanographic data was a collaborative effort between Old Dominion University and NCSA entitled *A Walk Through Chesapeake Bay* [Wheles et al.]. This *CAVE* application, first shown at the VROOM venue of the SIGGRAPH '94 computer graphics conference in Orlando, Florida, consisted of a three-dimensional graphical representation of a Chesapeake Bay bathymetry dataset textured and colored according to depth. Integrated into the virtual world were transparent isosurfaces derived from monthly composites of Chesapeake Bay surface salinity observations.

The bay research at Old Dominion focuses on seasonal effects on the marine life of the bay caused by pollutants and nutrients carried into the bay via the river systems and its interface with the Atlantic Ocean. The initial version of this application cycled over a full year. The user could activate the movement of time and simultaneously walk about the bay to explore the data. The passage of time was indicated by a clock icon that displayed one revolution per year, with each season identified by an icon on the clock. The user can observe the annual changes in salinity found in the bay due to increased freshwater runoff in the spring and heightened evaporation in the summer and fall. By peering just under the translucent sea surface, the user could view the main shipping channels and the abrupt topographic variations in the bathymetry.

An ambient soundtrack aided in achieving a sense of participant immersion. When the participant was above water,

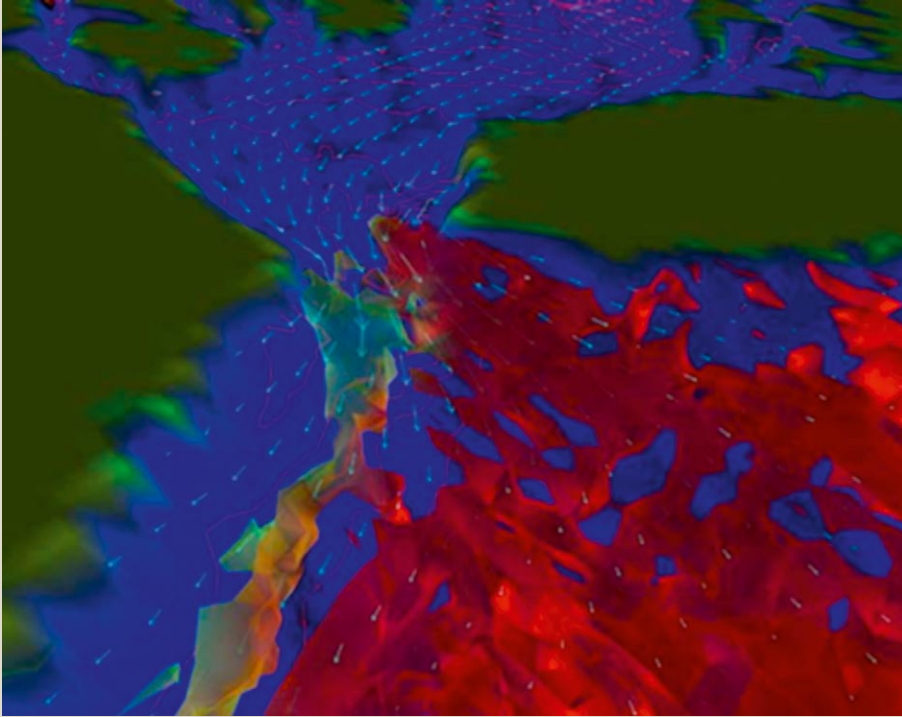


FIGURE 4-10

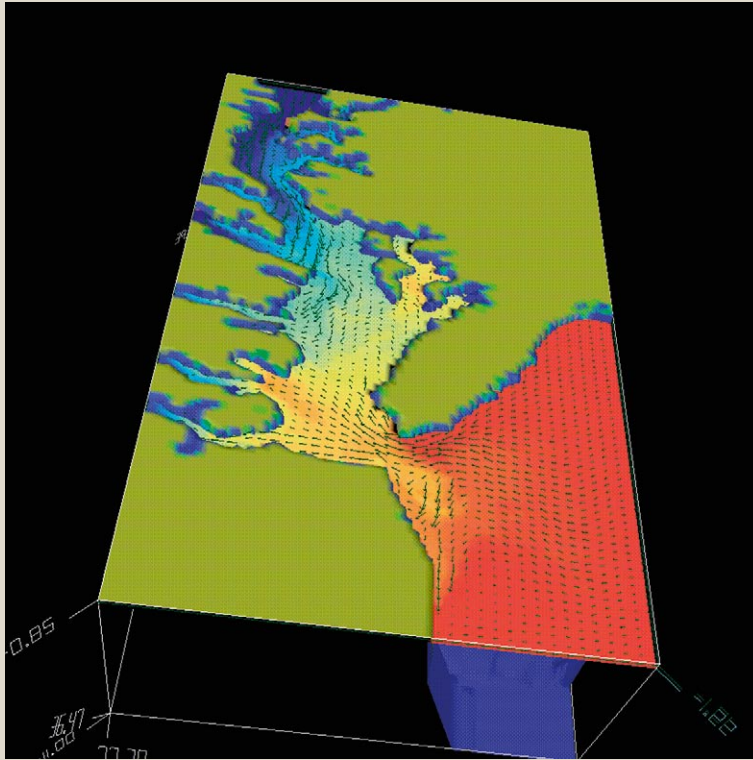
The Chesapeake Bay application provides a number of visual representations, including colored isosurfaces and vector arrows.

sea gulls and other beach sounds could be heard. Under water, bubbling and other underwater sounds helped the participant realize they were under the surface of the bay. A splash sound was made each time the participant crossed the boundary entering the water. When leaving the water, the participant heard the squawk of a gull indicating their return to the atmosphere.

In the initial prototype version of this experience, the virtual world consisted only of the bathymetry, salinity isosurfaces, and a few landmarks to indicate the geographic location of the bay. This allowed the researcher to examine and explain to others only some of the seasonal changes of the estuary system. Although the early prototype VR application was groundbreaking from both the standpoint of applying virtual reality to oceanographic information

as well as for viewing the annual salinity cycle in the Chesapeake Bay, it was obvious that more functionality was necessary. The next logical step was the development of a visualization framework to more easily integrate the results from *any* coupled physical-biological model into the virtual environment.

The application developers routinely use *CAVE5D* on their *ImmersaDesk* located in the Virtual Environments Laboratory at the Center for Coastal Physical Oceanography (CCPO) at Old Dominion University, where the freely available software is maintained and enhanced. It has been used to create virtual environments useful for the examination of the effects of flow on fish distributions in Beaufort Inlet, North Carolina [Wheless et al.], as well as the tidal circulation in Ponce Inlet, Florida.

**FIGURE 4-11**

Color, vectors, and depth provide details regarding the Chesapeake Bay in a virtual reality experience utilizing an ImmersaDesk.

Conclusion

In summary, this application works well as a demonstration of applied computational science. It is a good example which people are likely to understand because the content is familiar to them and can be used to tell an interesting and educational story. The public seemed to enjoy the notion of going for a virtual “walk under the bay.” From a science perspective, this application demonstrates great promise. The scientists involved have stated that this tool appears to be useful and beneficial to their research.

For the ODU *CAVE5D* project, the visualization files for *CBVE* provide the basis for the virtual environment and normally consist of a few hundred frames of multivariate data. These files routinely were as large as 450 Mbytes. Complete functionality in the networked interactive mode would

require that these visualization files be rendered at a sufficient frame rate (usually greater than 20/sec). If all graphics information is sent across the network, a sustained network throughput of much greater than 155 Mbits/sec would be required, which was too aggressive at the time. The commonality between the CEWES application and *CBVE* indicates that collaborative work between the sites would have enhanced both applications.

In addition to research on the Chesapeake Bay, other research institutions have used VR to visualize oceanographic simulations. Many of these institutions would be able to use some of the packages developed by ODU and CEWES in conjunction with NCSA. For example, Mississippi State University has done some interesting visualizations of water flow through the Sea of Japan using a FakeSpace

PUSH display, and in the NCSA *CAVE* [Gaither et al.]. They have now built their own *CAVE*-like system, which enables collaboration with other *CAVE* sites. By sharing work and combining efforts teams of researchers throughout the world will be able to create better tools for the specific purposes such as the oceanographic research.

Wheless says:

At present, the use of VR allows environmental scientists, educators, students and managers to

collaboratively view, analyze and interact with very large data sets in a way that is understandable and intuitive. By adding persistence to the shared virtual environment, so that the virtual space remains operating and changing independently of the users involved, we will enable worldwide participation and will allow raw data to become usable information and then cognitively useful knowledge.

CASE STUDY 4.3: NASA TELEPRESENCE

VEVI toolkit

The mere mention of the name “NASA” conjures images of high-tech equipment, adventure, and exploring new frontiers. And while NASA’s primary mission is the exploration of space, they put many new technologies to the test right here on Earth. Two technologies that NASA has used for exploring unfriendly terrestrial terrain are virtual reality and telepresence.

Historically, NASA has been at the forefront in developing applied virtual reality systems. Earlier in this chapter we looked at the *Virtual Windtunnel*, which is one of NASA’s uses of VR for the visualization of fluid flow around a solid body. The NASA Intelligent Mechanisms Group (IMG) is using telepresence and immersive displays to learn how to improve their ability to more efficiently control remote devices (e.g., robots).

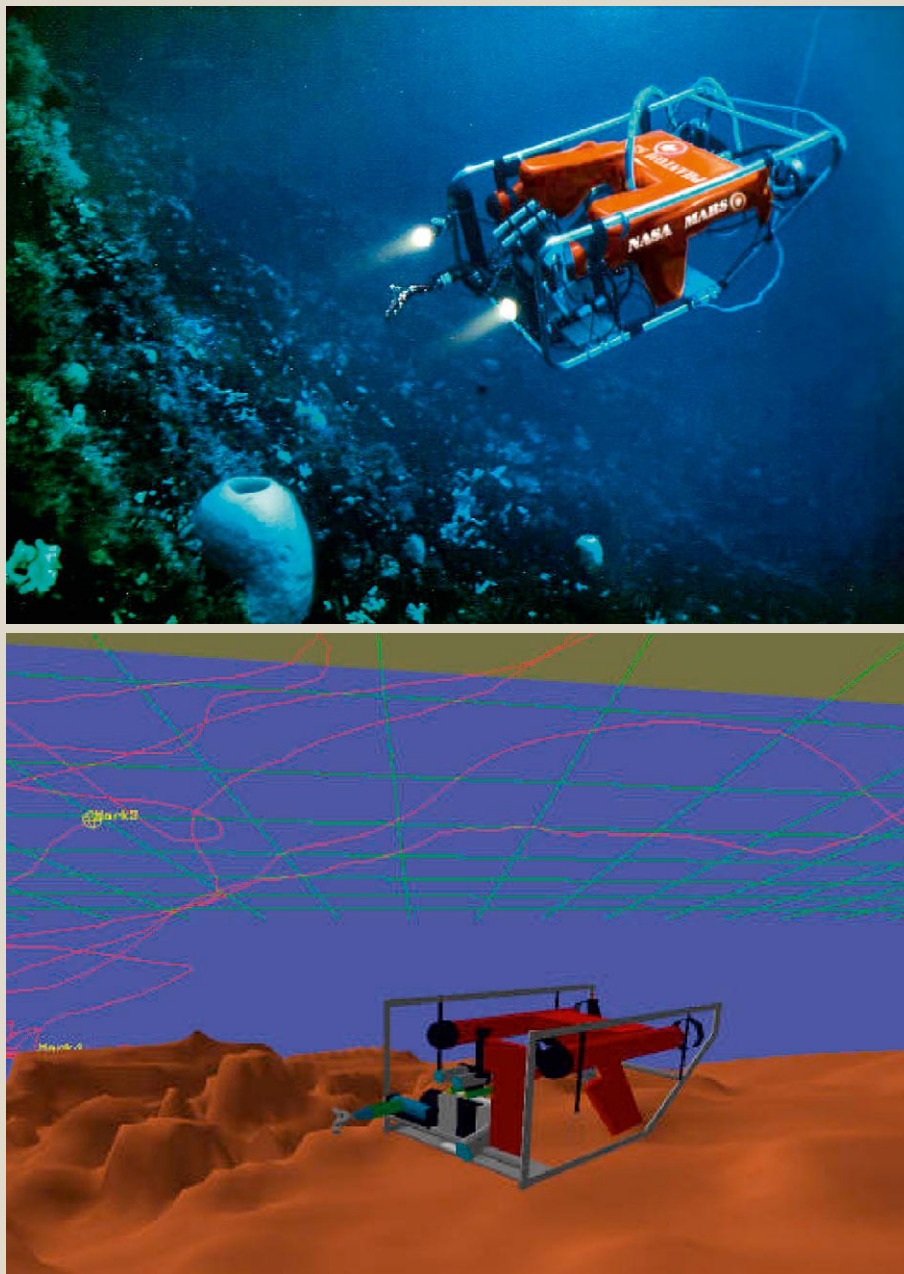
The application described in this section uses telepresence to allow scientists to view the depths of the ocean from a first-person point of view while remaining in safety on land or on a ship. The device they use can be operated either by using a typical computer system, or by using a tracked head-based display system that remotely controls the positioning of the device and cameras on the device in a natural way, by moving their head in a manner similar to what they would do if they were actually under the ocean. A specific focus of the IMG is to investigate technologies for the exploration of

planetary surfaces and space vehicle control. When testing these technologies, the preliminary stages of development are done right here on Earth, but often in hostile locations, including under the sea, in the desert, and in and around active volcanoes. Part of the research involved the creation of a software tool to control a variety of vehicles in a variety of situations. This tool is the *Virtual Environment Vehicle Interface (VEVI)*. This section focuses on the use of telepresence and other VR techniques employed in a particular mission of exploring the undersea world—one of the first where these elements were brought together at NASA.

Application Description

The first vehicle controlled by the *VEVI* system was the “Telepresence Remotely Operated Vehicle” (TROV). The TROV was designed as an undersea vehicle, where the buoyancy of the water acts as an analog to space vehicle travel. By putting the vehicle in a situation where real science is done, the system is put through a more realistic set of exercises. The TROV allowed scientists to explore, collect data, and chart the undersea world from the relative comfort and safety of the surface. The specific mission was to explore the undersea waters of the Ross Sea Ice near the McMurdo Science Station (McMurdo Sound) in Antarctica, including a benthic (ocean floor) ecological study of the region.

Much like a mission to another planet, the McMurdo Sound mission was a full-fledged scientific research project. The

**FIGURE 4-12**

An actual robot (top) explores the depths of the ocean. The operators are remotely located and control the robot via telepresence. A virtual re-creation of the ocean floor and the robot are seen below.

expedition leader, Dr. Carol Stoker, looked forward to discovering and cataloging the ecology of a previously unexplored undersea region. Other scientists involved with the mission include James Barry from the Monterey Bay Aquarium Research Institute (MBARI), James McClintock of the University of Alabama, and a team from the Antarctic Science Project S022. Barry was interested in learning how the dominant life-forms on the shelf vary as the depth increases, while McClintock planned to use the TROV to collect animal specimens to study their use of chemical defenses.

The TROV itself is an unmanned device for underwater exploration, remotely controlled by an operator at a significant distance. It has thruster motors to propel and steer the vehicle. Power, communication, and other electrical connections are supplied via a 340-meter tether from a console at the surface. An interface between the user and the remote location is provided via video cameras and a grasping arm mounted on the TROV. A satellite uplink connection allows the TROV to be controlled from NASA Ames Research Lab in California. Alternatively, a system at the dive site can be used.

Two control interfaces were used to interact with the TROV in this mission. The basic interface, which could be used at either site, consisted of a stereoscopic-capable computer display screen for output, and a keyboard, mouse, and joystick-equipped control box for input. The operator could view stereoscopic images and control the vehicle via the control box. No audio feedback was provided, apart from an intercom loop between the operator at Ames and the operator on the ice. This enabled the collaborators to coordinate activities, but did not provide any direct feedback from the interface with the TROV.

When controlled from NASA in California, another interface option was available—NASA's *Virtual Environment Vehicle Interface*. Using *VEVI*, the operator could view a stereoscopic computer monitor identical to the monitor used at the dive site, or use a head-tracked, stereoscopic, head-mounted VR interface.

The *VEVI* software system provides a user interface appropriate for different methods of teleoperative control. The primary methods available are supervisory control (infrequent task-level commands planned using *VEVI* and



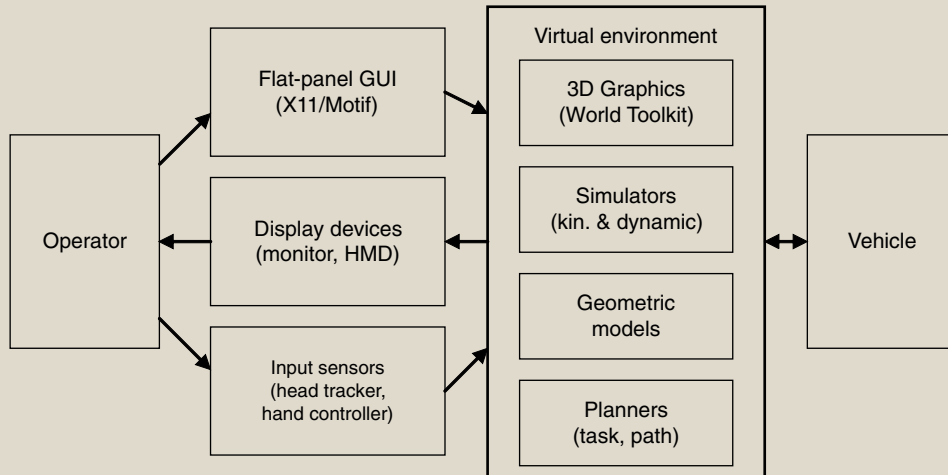
FIGURE 4-13

A user in California controlling the vehicle in the Antarctic Ocean. The immersive interface is provided by a VR2 HMD, Logitech Ultrasound tracker, and Spaceball.

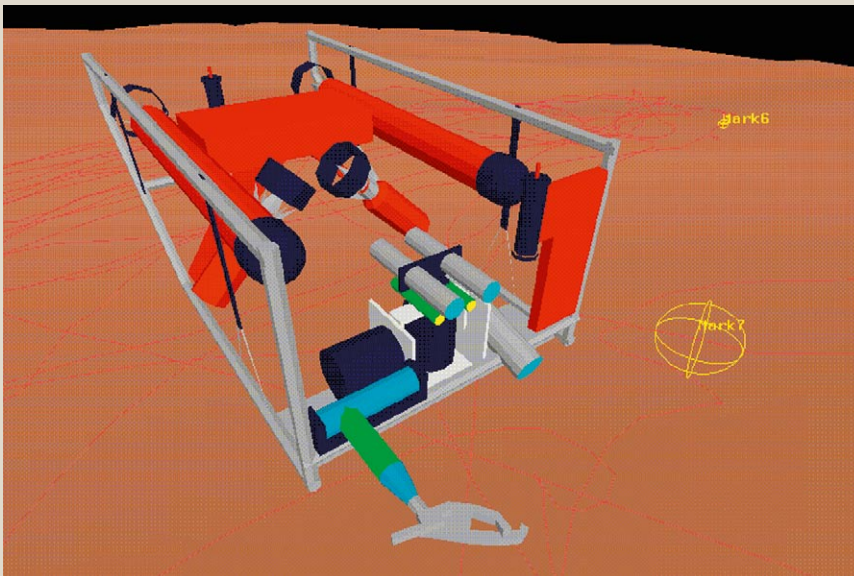
then sent to the vehicle to perform) and traded (or direct) control (commands and results are passed instantaneously between vehicle and operator) [Sheridan 1992].

The rendering system was independent of *VEVI*, and could be chosen by the mission programming team. For TROV, rendering was done using Sense8's *WorldToolKit (WTK)*. The goal was for the rendering to work adequately on a mid-range SGI (in 1994, they used an Indigo2 Extreme).

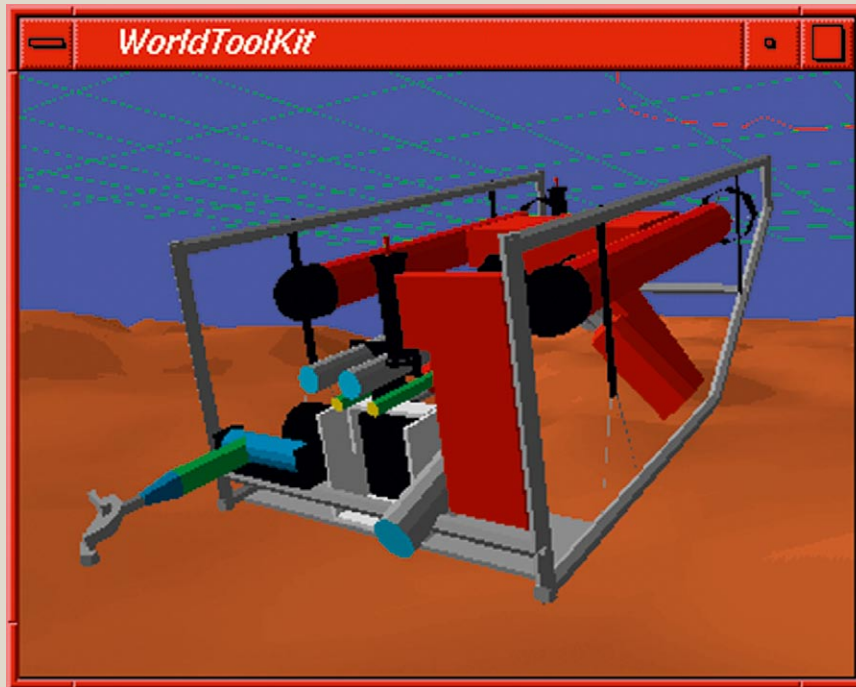
To test missions involving remote space operations, time delays can be artificially added to the communications to simulate the lag of space communication when doing terrestrial experiments. For example, whereas Moon communications take only 2.5 seconds for round-trip communications, communication between the Earth and Mars can take 4 to 40 minutes. More significant delays in communication force the usage of supervisory control methods. This feature has been utilized in the Nomad vehicle experiments conducted in the barren terrain of the Chilean Atacama Desert [Piguet et al.].

**FIGURE 4-14**

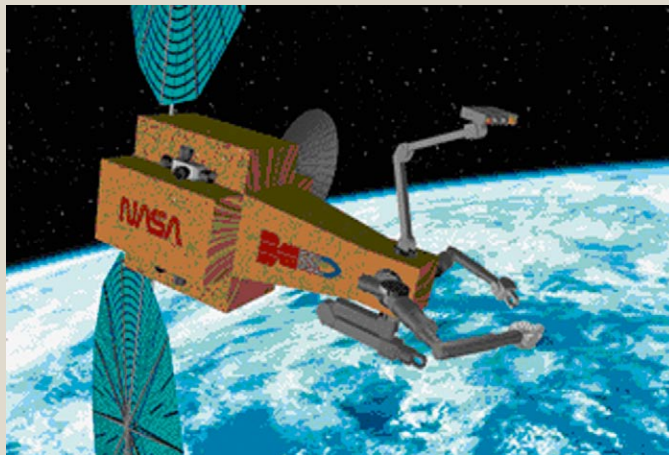
The system contains two layers of abstraction between the user and the vehicle. [Fong et al.]

**FIGURE 4-15**

The Sense8 WorldToolkit was chosen to provide the graphical rendering for the VEV system.

**FIGURE 4-15**

Continued

**FIGURE 4-16**

VEVI was used in controlling Ranger, an orbital robot used to repair satellites.



FIGURE 4-17

The Nomad planetary surface rover performed a 40 day mission to explore the Chilean Atacama Desert.

The VEV system provides two methods for representing remote locations. In the TROV McMurdo Sound mission, one method was to directly view the world via the live video cameras onboard the TROV. The other method was to use a computer-generated representation of the underwater terrain. The computer-generated representation was constructed using information gathered by a set of sensors carried on the TROV. A virtual-world model representing the real world was created by traversing the terrain with the TROV to gather the appropriate data.

The virtual-world display also included a representation of the position and state of the TROV, plus information such as the vehicle's track and science markers. Science markers are icons placed in the 3D world representing the location at which certain measurements were taken. The researcher can see the markers, to see where data was collected, and in later versions of the software, hyperlink to the raw or processed data. The data collected could be from any number of input sensors, resulting in a mixture of

data types, such as images, salinity values, temperature, or light measurement.

The two presentation methods give the operator a choice between viewing the remote locale via first- or second-person points of view. The first-person POV view is obtained from a pair of video cameras mounted on the vehicle. This technique includes separating the cameras by the distance common to human interocular spacing, and rapid tilt-and-pan movements to be able to follow quick human head rotations. Rapid response between human input and the visual feedback is an important factor in a telepresence application. The visual lag of a poorly responsive camera will cause perceptual difficulties for the operator. The stereoscopic camera-pair helps provide depth cues to the viewer, although since the cameras do not translate relative to the craft there is no depth cue from head-motion parallax.

In addition to the use of the stereoscopic camera-pair to produce a first-person view from the TROV, several other

**FIGURE 4-18**

The development team works on the Nomad planetary surface rover in the field.

cameras and sensors help to relay the current situation to the user. A monoscopic camera helps the user see what is immediately in front of the craft, and another monoscopic camera is mounted to the bottom of the craft to observe

what is directly beneath. The forward-looking camera is equipped with a power zoom lens, used to obtain close-up views. The vehicle also carries sensors for ambient light, direct light, dissolved oxygen, and temperature.

VR DISCOVERY

A significant difference between virtual reality and telepresence/AR applications is the increased demand for sensor devices. In order to allow interaction with a world, the interface must be able to accurately represent the world. When the world in question is the real world, the representation (and hence the ability to interact) relies on accurate sensing of the world plus the registration of the virtual-world representation to the real world.

When we directly view the real world around us with our eyes, we see it from a first-person point of view. The

computer generated version of the real world can also be presented from a first-person perspective. However, the computer version is also able to be presented as a second-person view of the world. Using this capability allows the operators to see “themselves” (i.e., the vehicle they are piloting) from an external point of view, giving them the ability to see the relationship between them and the environment.

Another important feature of telepresence is the ability to manipulate the world. Accordingly, the TROV was

(continued)

VR DISCOVERY (Continued)

equipped with a manipulator arm for gathering samples and executing other necessary activities in the ocean floor expedition. The TROV manipulator arm was furnished with a simple gripper claw to provide a mechanism by which the operator can affect the real world from within the virtual world.

The vehicle's position is tracked in much the same manner as ultrasonic and visual tracking systems are used to track the human body. Specifically the *SHARPS™* (by the Marquest Group) navigation system tracks the TROV by transmitting and receiving acoustic ranging signals. Transponders are mounted on the front and the rear of the vehicle. The transponders are referenced against three known position units. The fixed units are arranged in the water in the formation of an equilateral triangle. The transponders are positioned at a depth of 100 meters

(hung in the water by cables). Onshore, a 386-PC uses the tracking data to compute the vehicle's relative position from the dive hole from the tracking data. The PC also records the vehicle's movements, and keeps track of the sections of the sea covered by the vehicle's movements.

During the course of the McMurdo Sound benthic ecology project, the system was in nearly continuous use for over 2 months. The complete system was implemented with a variety of hardware, including Sun SPARCstations, Silicon Graphics systems and an assortment of special-purpose processing and communications boards. To aid in integrating this variety of platforms, the developers used NASA's *ARCA (Ames Robotic Computational Architecture)* standardized communications package for interprocess communication and synchronization of the heterogeneous, distributed system [Fong 1993].

VR System

The nonimmersive stereoscopic monitor display was implemented using CrystalEyes LCD shutter glasses. The VR display was implemented using a Virtual Research VR2 HMD. The HMD was tracked by a Logitech ultrasound position tracker. Both the immersive and nonimmersive systems used a Spaceball™ for input. The operators at the dive hole using the nonimmersive system also had a pair of joysticks on a lap-belt controller that they could use. In the immersive system, a head-tracking system provided user point-of-view input.

Vehicle movement was controlled using joysticks. Unlike some other telepresence applications, the manipulator arm was not controlled by a glove interface, but instead it was controlled using the joystick. The arm had limited possible configurations; thus it was sometimes necessary to manipulate the vehicle to achieve certain positions.

The terrain model is created by piloting the TROV along the sea floor while the system records data from the tracking system and onboard measuring devices. This data is then used to build the model. Once the terrain maps exist in the *VEVI* software, the vehicle can be driven purely within the virtual world, to an accuracy of .25 meters.

As with many explorative missions, multiple scientists were involved; each with their own goals to accomplish. One goal was to produce a video record of benthic (ocean floor) organisms gathered during passes of the TROV through several areas ranging from 60 to 1000 feet in depth. Another goal was to gather life-form samples from the region, and bring them back to the surface for further study. In addition to the marine scientists, the NASA engineers were interested in studying how VR and other displays could help the operator's ability to perform teleoperations.

To collect the samples, a separate basket constructed from PVC pipe and netting was created. A hinged door on the top of the basket opened by pushing on it. The scientist could explore by driving the TROV, finding samples, controlling the manipulator arm and claw with a joystick to procure a sample, bring it to the basket, and push it through the door. When the basket was full, they could grasp it in the claw and carry it to the surface with the TROV.

The developers feel that telepresence techniques provide significant advantages for studying the undersea world. Stereoscopic display was considered important in this application for several reasons. Stereoscopic imagery

helped provide a sense of depth that helped in using the manipulator arm. Also, the use of stereoscopic images provided a means to calculate the size of objects and the distance between objects. The navigation and tracking system also allowed the scientists to record measurements such as the distance traveled and the distance between objects.

This application shows an interesting integration of both telepresence and virtual reality techniques for use in the visualization of a remote environment, as well as controlling a vehicle within that environment. These techniques show great promise for exploring, analyzing, and manipulating in hostile, inaccessible, or extreme environments. Such techniques could feasibly be implemented in applications that require the control of vehicles in distant, dangerous, or extreme environments.

While the McMurdo Sound Project was focused on an under-sea environment, NASA has also done similar exploration on extraterrestrial environments. One of the key hurdles to overcome in such applications is the lag between the remote vehicle and the control operator. Even with sufficient bandwidth, the absolute distance creates a time delay that can lead to problems with conventional telepresence techniques.

The IMG researchers have made some interesting discoveries over the course of their research with this and other telepresence projects using the *VEVI* system. One thing they found was that once the move was made from the original text-based control systems to even the 2D interfaces, no one went back to the old method. The text method of reading enormous amounts of numbers and offering “indecipherable” command options was too “daunting.”

They also learned which interface features were key to aiding the understanding of the vehicle’s situation and how it could be controlled. Many of these features were added to the *VEVI* 2D interface for the descent of the *Dante* land vehicle into the rim of the Mt. Spurr volcano. In particular, they added visual reference aids such as grids, axes, landmarks, and orientation indicators to the display. The *VEVI* authors also found that improved registration between the real and virtual worlds greatly enhanced their ability to control the device. Many of these lessons are described in Fong’s paper on the *Dante* project [Fong et al. 1995]. Fong also found that “traded and shared supervisory control can improve system

performance through synergistic human-machine interaction,” and that user situational awareness and information processing can be enhanced via a virtual reality interface.

Butler Hine, Group Leader of the Intelligent Mechanisms Group during the TROV and *Dante II* missions, felt that the *Dante II* mission was where using VR as part of the operator interface resulted in a significant improvement in usability. In many ways, TROV was “basically a telepresence device retrofitted to use a VR interface. It lacked the on-board control system and sensors required to really do a good job in the VR interface.” Hine also stated that “*Dante II*, on the other hand, was very sensor rich and had a high-level control system. The vehicle was complex enough that the gain in situation awareness from using a VR interface was substantial.”

One feature that had not been used in many NASA telepresence missions was the use of audio feedback, either as part of the control interface or to hear sounds from the remote location. In one mission to Mt. Kilauea, an active volcano in Hawaii, an audio pickup was placed on the rover, providing sounds as well as sights back to the operators. Hine points out that “it is amazing how good the human ear is at determining the health of a system by sound alone,” indicating the value of being able to hear as well as see the telelocation.

Related Applications and Follow-on Work

VEVI was used in the *Pathfinder* mission to Mars, but as a visualization tool rather than a control tool. Terrain data received from *Pathfinder* was displayed and validated using *VEVI*. Control operations for *Pathfinder* were planned out a day ahead of time and sent as a mission for *Sojourner* (the Mars rover component of the mission). The planned operations were created using a separate, custom, tool also developed by the NASA IMG. The daily preplanned operating method did not require many of the features of *VEVI*, such as real-time telemetry display, or real-time control of the vehicle.

Conclusion

In addition to Butler Hine, the architects and implementers of the *VEVI* system include Laurent Pigué, Terry Fong, Erik Nygren, and Phil Hontalas. This team collaborated with many other people when integrating *VEVI* for the hardware of a specific mission. Some of the robotic vehicles were made by the Carnegie Mellon Robotics Institute, including the *Dante II* and *Nomad* vehicles.

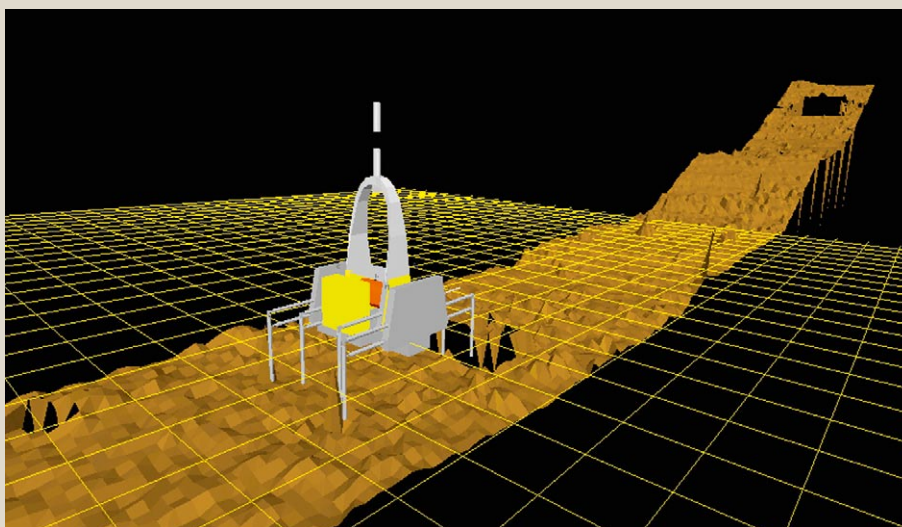


FIGURE 4-19
The actual system and its virtual representation.



FIGURE 4-20

Operators utilize both conventional video and virtual representation for remote control of the system.

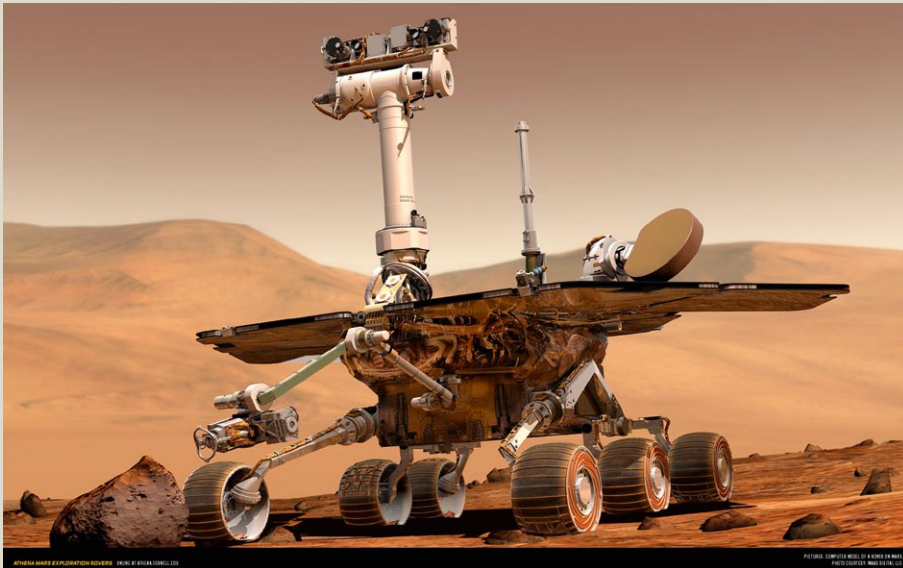


FIGURE 4-21

In addition to real-time control, applications for virtual reality include visualization of remote environments and reconstruction of explored spaces through collected data.

Many of the IMG researchers responsible for the VEV system have gone on to found a company to extend and support VEV, as well as produce new technologies for telepresence applications. The name of their company is Fourth Planet Inc. Butler Hine, now president and CEO of Fourth Planet, seeing a great need for putting this type of technology to use, states:

I think that virtual reality and telepresence can have an enormous impact on how we explore and work in remote hazardous locations. These techniques will ultimately give us the ability to send our best minds and talent to places which would otherwise involve prohibitively extreme physical risk.

CASE STUDY 4.4: GRIP

Molecular Visualization

The University of North Carolina at Chapel Hill has used virtual reality to investigate interactions between molecules, such as drug-protein interaction using virtual reality. A significant feature of their system is the use of a six-degree-of-freedom force-feedback device called the Argonne Remote Manipulator (ARM). The ARM allows the scientist to “feel” the forces between two molecules as they are placed in different locations and orientations while using the ARM also as an input device.

One could consider this set of applications to be as much about haptic feedback as it is about molecular visualization, the fact that a significant investment in investigating the use of haptic devices makes this virtual reality application stand out from the pack of other molecular visualization application. It is very interesting to use the ARM to place the molecules in different orientations and “feel” when the molecules are attracting or repelling each other.

Application Description

Most of the applications described in this book rely on the visual sense to convey a virtual world to the human participating in the world. Many of these also use sounds to enhance or augment the information conveyed visually. However, very few use the sense of touch, or in this case more specifically kinesthesia, to transmit information about the world to the participant. (See Chapter 5 for an application that uses force feedback for surgical training.) The GROPE series of projects are significant in that not only

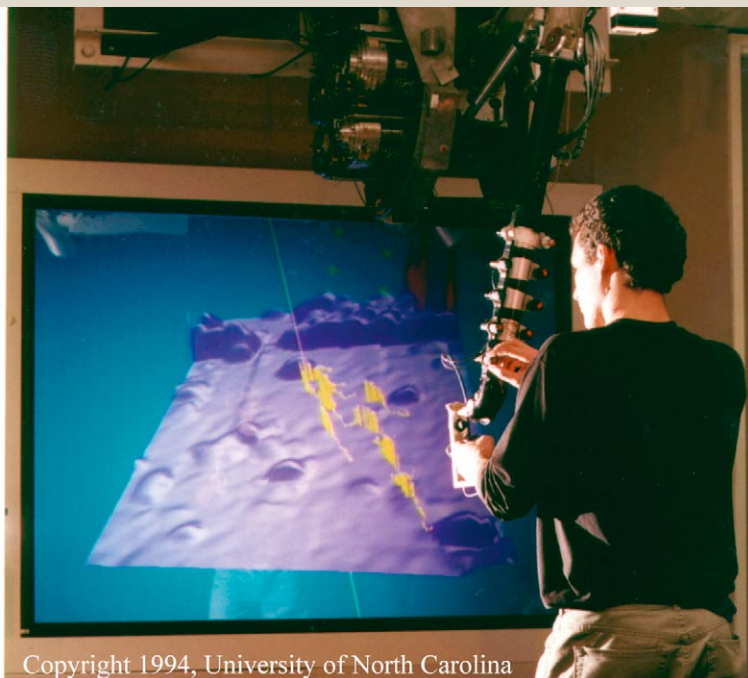
are they one of the few uses of haptic display for scientific visualization, but that this research has been ongoing at the University of North Carolina at Chapel Hill (UNC) since 1967 [Brooks et al.].

Inspired by Ivan Sutherland’s description of *The Ultimate Display*, the UNC Department of Computer Science set out not only to implement such a device, including haptics, which was also part of Sutherland’s vision, but also to put this ultimate display to work in aiding with real-world tasks. Founding department head and Computer Science pioneer Fred Brooks spearheaded the effort to create the new technology in such a way that it would be demonstrably useful.

Significantly, this project has been as much about analyzing the benefits of haptic technology as much as it has been about producing a single application that was useful to a particular set of scientists. The UNC research team has conducted a series of experiments on the usefulness of the concepts of the GROPE project while technology has allowed the application to advance.

From the outset of the project, the research team led by Brooks strove to implement an application for interactive exploration of molecular docking. Until *GROPE III* the research focused on taking steps that would take them in the direction of the originally conceived molecular docking application. The initial experiments focused on much simpler tasks.

Their research project has expanded into other application areas. The overall research for molecular “visualization”



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FIGURE 4-22

The ARM provides six-degree-of-freedom force feedback of interactions in the virtual world.

at UNC is now referred to as *GRIP* (*GRaphical Interaction with Proteins*), with the *GROPE* project and others falling within. Of the *GRIP* projects, *GROPE* and *nanoManipulator* use haptics as a means of perception and interaction.

The generalized result of their data indicates that haptic display increases performance of a molecular docking task by about twofold in the *GROPE* task, and the *nanoManipulator* project has anecdotally demonstrated tasks that can only be performed with the haptic interface.

The *GROPE* project had evolved through many generations to become a software application usable by a number of chemists at multiple facilities. The resultant application (*GROPE III*) still provides for the same basic operation—finding the best docking configuration between a drug and a protein—while offering a variety of representational idioms to aid in this task.

Although “scientific visualizations” often map a variety of data to the visual sense (and to a lesser degree the aural sense) typically only simulated forces are mapped to our kinesthetic sense. This phenomenon is probably mostly due to the limited number of haptic displays that are in use. Thus, the primary owners of such systems are groups that see a direct need for perceiving forces. In the case of the *GROPE* project, the forces displayed to the user are the natural attraction and repulsion forces between molecules.

The molecular docking task was chosen because it involves real-world forces and torques that need to be analyzed by chemists working on drug design. While mapping a force phenomenon to the force display, the *GROPE* system must map the force “across scale.” That is, it represents a force that is very common, but not otherwise perceptible at human scale. Bringing the drug interaction phenomenon to the level with which a human researcher

can interact on their terms allows them to get a better “feel” for the chemical process by exploring it directly.

A variety of representational choices are provided to the user of the *GROPE III* system. The representation of each molecule can be selected separately from a choice of none (i.e., invisible), ball and stick, stick only, as well as a Connolly dot surface to represent the active site.

The forces between the drug and protein can be represented in one of two fundamental ways. The force can be represented visually, using an arrow that points in the direction of the force, with the length of the arrow proportionate to the magnitude of the force. The force can also be rendered haptically, producing directional forces on the user's hand or finger on all types of haptic display, and rotational forces on those capable of full 6DOF force rendering.

Two graphical-only representations in the *GROPE III* application include energy “thermometers” that show the total energy and the internal energy of the drug molecule, and a vector representation that shows forces and torques as arrows, with the force arrow emanating from the point, and the torque arrow shown as a tangent to a sphere surrounding the point. When subjects were only given the visual representation of the force and torque, they used the non-optimal strategy of attempting to first minimize the force and then the torque, versus doing both at the same time as a single operation. Those subjects interacting with the force display generally minimized both at the same time.

VR System

Development of the *GROPE* system stepped through some specific advances in force and computational technology over the years. These steps can be categorized as four distinct stages:

- 2DOF force system (called *GROPE I*, 1971)
- 3DOF force system with a simple task (*GROPE II*, 1976)
- 6DOF force system with a simple task (*GROPE IIIA*, 1989)
- 6DOF force system with a 6D molecular docking task (*GROPE IIIB*, 1990)

Because force feedback is not a commonly sought form of sensory display, a major problem with experimenting with

force rendering systems was simply acquiring the hardware. This was particularly true in 1967. The main source of haptic display devices at that period were produced for teleoperation devices. UNC managed to acquire an orphaned Argonne Remote Manipulator (ARM) teleoperator unit.

As better force displays and computational systems became available, the task stepped closer to the original goal of the project. The initial 2DOF application (*GROPE I*) used potentiometers and servos to sense and move a small knob within a 2-inch by 2-inch square. The application was an experiment to measure how well people could judge a continuous 2D force displayed on this device. The *GROPE II* application used the ARM device as one means of interacting with a simple block world. Subjects were asked to make comparisons of particular cues in the virtual world using force, perspective, stereoscopic, and other representational cues about the world.

As the results of the *GROPE II* project were published, Brooks and the research team decided that computers would have to improve by 100-fold before full 6DOF forces could be calculated in real time. By 1986, this threshold had been reached, and the *GROPE* project was reborn as *GROPE III*.

The *GROPE III* system also used the ARM haptic device, but with more powerful computational and rendering computers. Computers were now powerful enough to calculate a 6DOF force in a simple world at about 80Hz (which was determined to be the rate above which people could not sense any difference on the ARM system). The world was rendered on an Evans and Sutherland PS-330 color vector display, and computed on a dedicated Sun4/280 UNIX workstation.

Brooks' research team hoped to answer several questions about the potential usefulness of haptic displays in a scientific visualization task. The primary questions were:

- Do haptic displays demonstrably aid perception?
- How big is the effect?
- What models are best suited to force display?
- How good does the display need to be?
- Where is the perception threshold above which there is little/no gain?

In *GROPE I* the major finding was that people could learn from the system, but that if they weren't interested in the task, there was no gain in knowledge.

In the *GROPE II* research experiment, the team found that while each individual relies on different cues to discern the three dimensional nature of shapes, there are cues that are stronger than others. Specifically, they found that force feedback (touch) provides a better perceptual cue regarding the shape of objects in the world than either stereoscopic images or motion parallax. They found that force cue was somewhat weaker than the cue provided by shadows. Another finding was that when the user interface mediates 3D movement with 2D widgets, then people tend to limit their work to 2-DOF at a time. They separate the orientation portion of maneuvering from location, making the task less holistic.

In *GROPE III*, where the experimental task approached the original concept of using force as an interface to a molecular docking task, they found that adding haptic augmentation to the interface improved the results by a factor of two. Chemists who have used *GROPE III* have demonstrated that they are able to find good docking positions with it. More importantly, these chemists report that they acquired a better understanding of the details of the drug receptor site by experiencing the forces at work on the docking molecules. Thus in general, the UNC developers have shown haptic interfaces to have a potential benefit in a variety of computer applications.

Related Applications and Follow-on Work

Another project in the *GRIP* domain that uses haptic display is the *nanoManipulator*. The *nanoManipulator* project was begun in 1991 to use force feedback devices as an interface to scanned-probe microscopes (SPM). SPMs operate by scanning the surface of a material at the atomic level. The probe of these microscopes can also be used to modify the surface. By using force displays as an interface to these devices, users can both feel and affect the object under purview.

As with the *GROPE* applications, different force feedback devices can be used to interact with the *nanoManipulator*. However, they typically use a SensAble Technologies PHANToM device (a 3DOF point force display). Unlike *GROPE*, the *nanoManipulator* can use head tracking to

visually immerse the user as well as haptically immerse them. They have used both workbench-style visual displays with and without head-tracking, and HMD visual displays.

Because the *nanoManipulator* generally uses live, real-time data, it is as much a telepresence application as it is virtual reality. In effect, what it is doing is allowing the user to jump from the world around them to the atomic world near the tip of the microscope, at one-millionth the scale. The real-world information can then be augmented with colors, forces, or other information to enhance the user's perception of the world.

With the *nanoManipulator* application, the UNC researchers found that the haptic VR interface enabled insights that were not made by scientists who had worked with the same data using traditional methods. Of course, this does not imply that the previous visualization methods should be abandoned.

One group they had collaborated with discovered a feature of their own data ("graphite sheets coming out of the surface") that they had not noticed using their traditional methodology—pseudo-color imaging [Taylor et al., 1997]. The spectral highlights caused by viewing the surface from a particular angle caused the feature to become apparent while flying over the surface. On the other hand, there is often a substantial amount of noise in the SPM data that can become prominent in 3D renderings and thus obscure features that are readily evident in the traditional 2D pseudo-color view. Both the 2D and 3D views are made available to the user to allow information to be gathered using the most appropriate view for a particular task.

The graphite sheet anecdote speaks favorably for using head tracking for visual displays, but such use could not make use of the haptic interface provided by the *nanoManipulator*. The primary benefits afforded by the *nanoManipulator* from the haptic interface device are in the improved user interface it provides with the scientific instruments. Without the *nanoManipulator* acting as mediator between the user and the instrument, the user has limited options for interacting with the surface. Two new features that emerge from the computer mediation of the device are extending how the tip can be used to interact with the surface, allowing the user to feel the material during times when the visual scan is unavailable.

Normally, the user of an SPM has only the brute force method of using the microscope tip to interact with the material. This brute force interaction is like probing and manipulating a block of ice with an ice pick. The *nanoManipulator* adds the ability to interact with the surface in a variety of ways that are more appropriate for certain tasks. These “virtual” methods of tip control include a “broom” tip and a “sewing” tip. The broom tip allows nanotubes to be “swept” across the surface by rapidly moving the tip back and forth between the two ends of the “broom.” Specifying the two end points of the broom is a far easier task for the user than trying to move the tip back and forth on their own. Another example of a “virtual tip” is the “sewing” tip. This tip is used when the user wishes to create a “trench” in the surface of the material, but doesn’t want the tip to plow through the material and cause a “snow plow” ridge to form. To prevent material from building up, the tip presses into the surface, then rises up, moves a little, and then presses down again in rapid succession, much like a sewing machine needle.

The other major benefit that arises from computer mediation between the user and the “remote” environment is that the haptic display can continue to allow the user to perceive the material even when the visual display cannot be updated. Because of how scanning-probe microscopes work, they can either use the probe to scan the surface of the material or to interact with the surface. Thus, when interacting, the standard (non-computer-mediated) interface is to operate “blind.” The visual display can provide an image from the most recent scan, but it won’t show any changes that the user is making to the material. Also, because doing a scan on the surface leaves the microscope tip in an indeterminate location, the user cannot request a scan in the middle of a manipulation and expect to continue where they left off.

Operations that might otherwise be impossible to perform, such as sliding a particle across a surface to fill a gap in a wire, can now be done. The user cannot see when the particle is slipping from the front of the tip, but they can feel it and readjust their movement accordingly. Another example is that the depression between two carbon filaments can be sensed allowing the user to “tease” them apart [Taylor et al.].

In addition to allowing users to perceive the microscopic world during times when visual representation is not possible, haptic feedback can also provide perceptual cues that are masked by the visual representation. For example, adenovirus particles appear roundish in the visual display, but the discrete edges can be felt with the haptic interface.

Conclusion

Although many people assume that virtual reality is about visual imagery, and sometimes sounds, this application shows a cogent use of haptic representation of information. By integrating the participants’ bodies in a way that allows them to feel the intermolecular forces of the subject of study, they are able to gain another channel of information to assist them in gaining insight into the inner workings of the virtual world they are experiencing. As a tool for science, the *nanoManipulator* allows data to be stored and retrieved. This is done because scientific research must be reviewable in order to be helpful. As Fred Brooks states as a general principle of scientific research: “If it isn’t in the lab notebook, it didn’t happen.” In other words, “an experiment is performed once, but may be analyzed many times.” In the case of the *nanoManipulator*, the “notebook” can store the entire history of a user’s interaction with a material—both the scan data and the user modifications or touching of the surface. Researchers have already had occurrences where they saw things during the replay that hadn’t been noticed during the experiment.

In general, the *nanoManipulator* demonstrates that by mediating the interface between user and instrument, the VR interface increases the usefulness of the instrument itself. In turn, the virtual reality application becomes a valuable tool for scientific exploration. At the same time however, it is important to keep in mind the lesson the *nanoManipulator* team has learned that the addition of new representations and interfaces does not mean that the previous methods are not valuable and should be discarded. Often there are features that are still best detected with the old method, and by switching back and forth between representations, the scientist can make a better overall mental model of what is being presented to them.

A number of people have worked on the *GRIP* research projects. Professor Fred Brooks is the overall project lead, and he conceived the drug-docking application. The *nanoManipulator*

project was conceived by Warren Robinett and Stan Williams, and included Professor Russell Taylor, Richard Superfine, and Sean Washbur. The University of North Carolina research was funded by the National Institute of Health/National Center for Research Resources (NIH/RCRR) and by the National Science Foundation (NSF) in support of these efforts.

The *GRIP* projects are interesting both because there is anecdotal evidence that they have benefited some scientists and, more importantly, because there is experimental

evidence that the haptic interfaces can improve performance with the virtual or teleworlds.

Other sites have begun to use the *GROPE III* software for their own molecular docking research. In particular, the Wright-Patterson Air Force Base, working with the National Center for Supercomputing Applications (NCSA) and Russell Taylor at UNC, has evolved the software to use more visual representations of the molecular systems, use the PHANToM device, and run in the *CAVE* environment.